

vLLM Evolution Brief

Official-release evolution through 2026-09-19 · latest stable: v0.29.0

One-line conclusion

vLLM evolved through release-documented engine, scheduling, model-support, and distributed-serving changes. This brief only summarizes official-source evidence.

Release evolution timeline

Stage	Representative releases	Architectural transition
Foundation	v0.1.0 – v0.3.3	paged KV management, continuous batching, early quantization, Multi-LoRA, prefix caching, and FP8 KV cache.
Production engine	v0.4.0 – v0.6.6.post1	automatic prefix caching, chunked prefill, multimodal serving, multi-step scheduling, and asynchronous output.
V1 transition	v0.7.0 – v0.10.2	V1 alpha/default, torch.compile, DeepSeek MLA/EP/DP, Blackwell, and disaggregated prefill.
V1-only scaling	v0.11.0 – v0.13.0	V0 removal, CUDA graphs, CPU KV offload, Model Runner V2 preview, DBO, and disaggregation.
Async and heterogeneous execution	v0.14.0 – v0.20.2	async scheduling, MRv2 maturation, FA4, hybrid cache, realtime/gRPC APIs, elastic EP, and vLLM IR.
MRv2 platform	v0.21.0 – v0.25.1	MRv2 defaults, Rust frontend, multi-tier KV, DeepEP v2, universal speculative decoding, and legacy PagedAttention implementation removal.

Engine path · scale path · model path

Engine: release-documented execution, compilation, and kernel changes. Scale: release-documented parallelism and disaggregation changes. Models: release-documented dense, MoE, multimodal, speech, and hybrid-model support. Consult configured official documentation for current architecture details.

Key version milestones

Date	Release	Official source
2023-06-20	v0.1.0	https://github.com/vllm-project/vllm/releases/tag/v0.1.0
2023-10-16	v0.2.1	https://github.com/vllm-project/vllm/releases/tag/v0.2.1
2024-03-01	v0.3.3	https://github.com/vllm-project/vllm/releases/tag/v0.3.3
2024-08-05	v0.5.4	https://github.com/vllm-project/vllm/releases/tag/v0.5.4
2025-02-01	v0.7.1	https://github.com/vllm-project/vllm/releases/tag/v0.7.1
2025-05-15	v0.9.0	https://github.com/vllm-project/vllm/releases/tag/v0.9.0
2025-11-20	v0.11.2	https://github.com/vllm-project/vllm/releases/tag/v0.11.2
2026-03-11	v0.17.1	https://github.com/vllm-project/vllm/releases/tag/v0.17.1
2026-06-05	v0.22.1	https://github.com/vllm-project/vllm/releases/tag/v0.22.1
2026-09-09	v0.29.0	https://github.com/vllm-project/vllm/releases/tag/v0.29.0

Deployment takeaways

Why vLLM: release-documented serving and scaling evolution. What to manage: version cadence, model/hardware compatibility, cache and parallel settings, and migration notes for defaults/deprecations.

Sources: official GitHub Releases and configured official documentation. Cutoff 2026-09-19; source mode cache.